

Aggregate Demand I: Building the *IS-LM* Model

Chapter 13

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

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In This Chapter, You Will Learn:

About the IS curve and its relationship to:

- ▶ the Keynesian cross
- ▶ the loanable funds model

About the LM curve and its relationship to:

- ▶ the theory of liquidity preference

How the $IS-LM$ model determines income and the interest rate in the short run when P is fixed.

Context, part 1

- ▶ Chapter 12 introduced the model of aggregate demand and aggregate supply.
- ▶ *Long run*:
 - ▶ prices flexible
 - ▶ output determined by factors of production and technology
 - ▶ unemployment equals its natural rate
- ▶ *Short run*:
 - ▶ prices fixed
 - ▶ output determined by aggregate demand
 - ▶ unemployment negatively related to output

Context, part 2

- ▶ This chapter develops the $IS-LM$ model, the basis of the aggregate demand curve.
- ▶ We focus on the short run and assume the price level is fixed (so the $SRAS$ curve is horizontal).
- ▶ Chapters 13 and 14 focus on the closed-economy case.
- ▶ Chapter 15 presents the open-economy case.

The Keynesian cross

- ▶ A simple closed-economy model in which income is determined by expenditure.
(due to J.M. Keynes)
- ▶ Notation:
 I = planned investment
 $PE = C + I + G$ = planned expenditure
 Y = real GDP = actual expenditure
- ▶ Difference between actual and planned expenditure = unplanned inventory investment

Elements of the Keynesian cross

Consumption function: $C = C(Y - T)$

Government policy variables: $G = \bar{G}$, $T = \bar{T}$

For now, planned investment is exogenous: $I = \bar{I}$

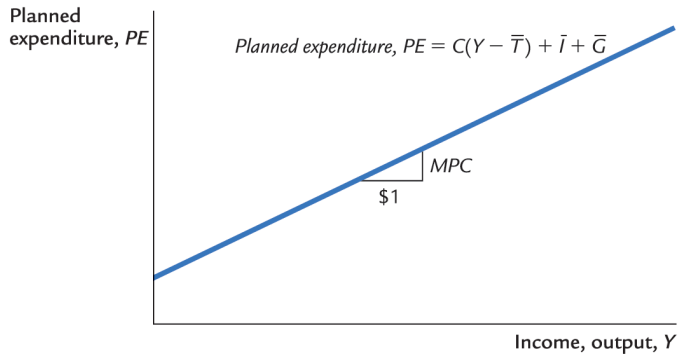
Planned expenditure: $PE = C(Y - \bar{T}) + \bar{I} + \bar{G}$

Equilibrium condition:

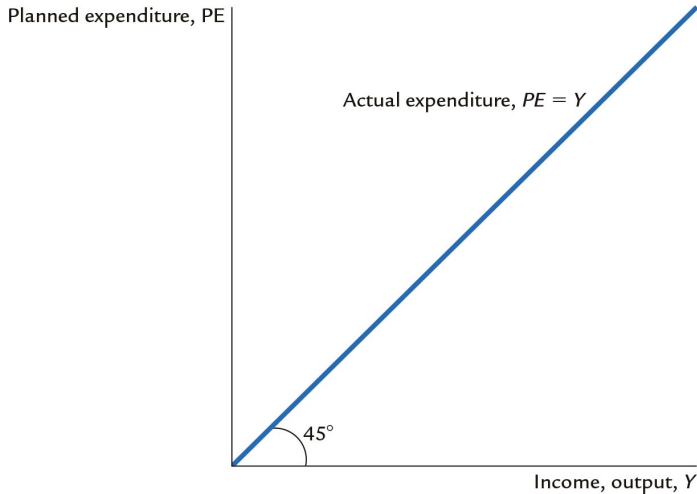
Actual expenditure = planned expenditure

$$Y = PE$$

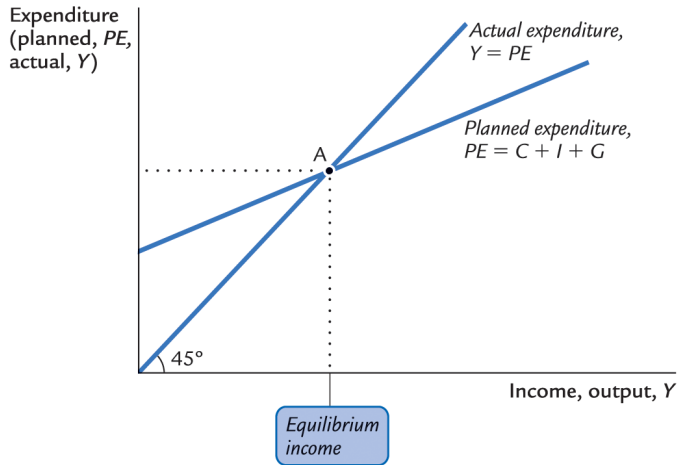
Graphing planned expenditure



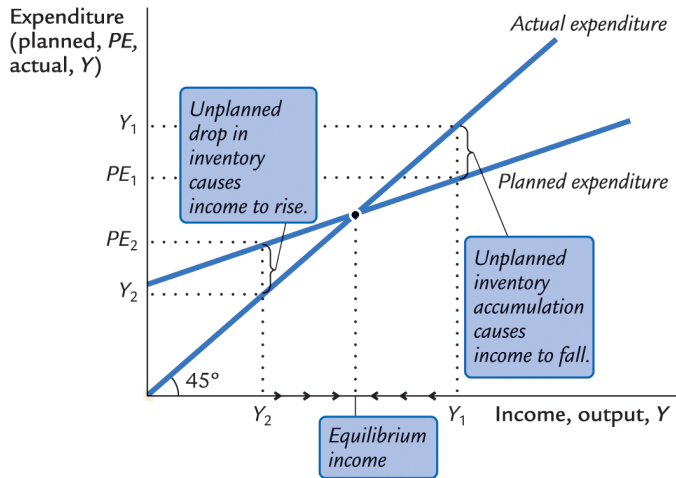
Graphing the equilibrium condition



The equilibrium value of income



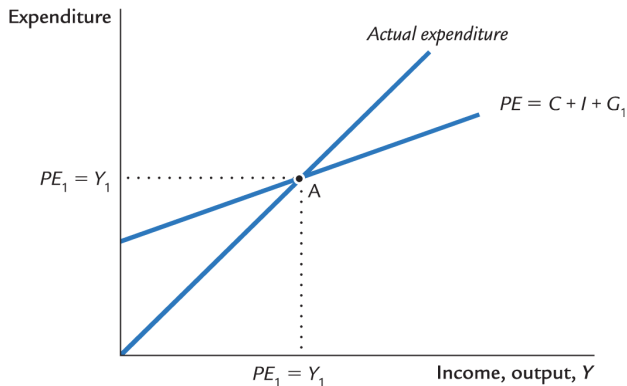
The adjustment to equilibrium



An increase in government purchases

An increase in G increases PE by ΔG . Equilibrium moves from A to B ; income rises by more than the increase in G .

The initial increase in G increases Y . Given $C = C(Y - T)$, higher income causes higher consumption. The multiplier process continues until the new equilibrium (Y_2) is reached.



Solving for ΔY (1 of 2)

$Y = C + I + G$ equilibrium condition

$\Delta Y = \Delta C + \Delta I + \Delta G$ in changes

$= \Delta C + \Delta G$ as I is fixed

$= (MPC \times \Delta Y) + \Delta G$ as $\Delta C = MPC \times \Delta Y$

Collect terms with ΔY on the left side:

$$(1 - MPC) \times \Delta Y = \Delta G$$

Solve for ΔY :

$$\Delta Y = \frac{1}{1 - MPC} \times \Delta G$$

The government purchases multiplier

Definition: the increase in income resulting from a \$1 increase in G .

In this model, the government purchases multiplier equals

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - MPC}$$

Example: If $MPC = 0.8$, then

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - 0.8} = 5$$

An increase in G causes income to increase 5 times as much!

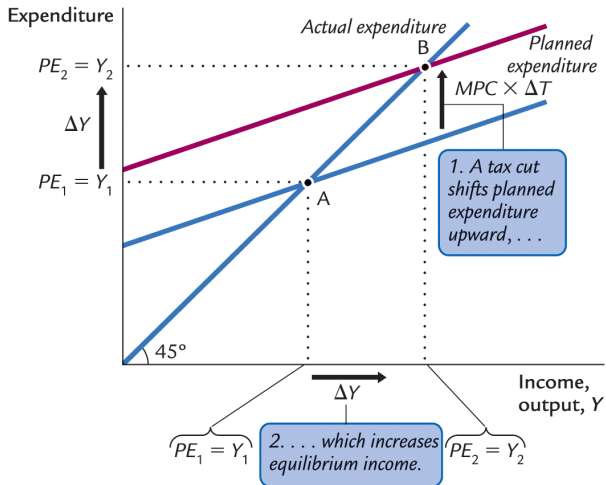
► Full IS-LM example

Why the multiplier is greater than 1

- ▶ Initially, the increase in G causes an equal increase in Y : $\Delta Y = \Delta G$.
- ▶ But $\uparrow Y$ causes $\uparrow C$
 - which causes further $\uparrow Y$
 - which then causes further $\uparrow C$
 - which then causes further $\uparrow Y$
- ▶ So the final impact on income is much bigger than the initial ΔG .

An increase in taxes

A decrease in taxes increases planned expenditure (PE) by $MPC \times \Delta T$. Equilibrium moves from A to B ; assuming $MPC > 0.5$, income rises by more than the decrease in T .



Solving for ΔY (2 of 2)

$Y = C + I + G$ equilibrium condition

$\Delta Y = \Delta C + \Delta I + \Delta G$ in changes

$= \Delta C$ as G and I are fixed

$= MPC \times (\Delta Y - \Delta T)$ as $\Delta C = MPC \times (\Delta Y - \Delta T)$

Solving for ΔY : $(1 - MPC) \times \Delta Y = -MPC \times \Delta T$

Final result:

$$\Delta Y = \frac{-MPC}{1 - MPC} \times \Delta T$$

The tax multiplier, part 1

Definition: the change in income resulting from a \$1 increase in T :

$$\frac{\Delta Y}{\Delta T} = \frac{-MPC}{1 - MPC}$$

If $MPC = 0.8$, then the tax multiplier equals

$$\frac{\Delta Y}{\Delta T} = \frac{-0.8}{1 - 0.8} = \frac{-0.8}{0.2} = -4$$

The tax multiplier, part 2

...is *negative*:

A tax increase reduces C , which reduces income.

...is *greater than one* (in absolute value) if $MPC > 0.5$:

A change in taxes has a multiplier effect on income.

...is *smaller than the government spending multiplier*:

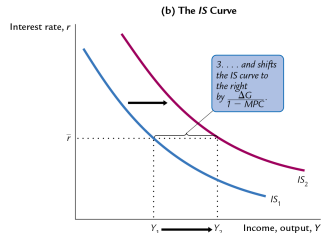
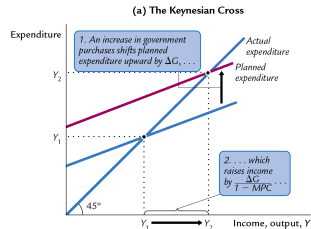
Consumers save the fraction $(1 - MPC)$ of a tax cut, so the initial boost in spending from a tax cut is smaller than from an equal increase in G .

NOW YOU TRY: Practice with the Keynesian cross

Use a graph of the Keynesian cross to show the effects of an increase in planned investment on the equilibrium level of income/output.

NOW YOU TRY: Practice with the Keynesian cross, answer

An increase in planned investment increases planned expenditure (PE) by ΔI . Equilibrium moves from A to B .



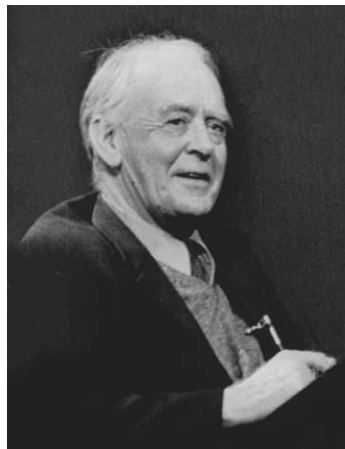
The *IS* curve

Definition: a graph of all combinations of r and Y that result in goods market equilibrium

For example: actual expenditure (output) = planned expenditure

The equation for the *IS* curve is:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$



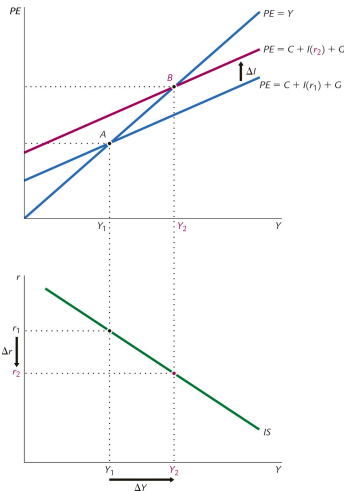
Sir John R. Hicks
(1904–1989)

formalized *IS*–*LM*
in 1937.

Deriving the *IS* curve

The real interest rate decreases from r_1 to r_2 , causing planned investment to increase. The increase in planned investment increases income from Y_1 to Y_2 .

The *IS* curve summarizes the relationship: the lower the interest rate, the higher the level of income.



Why the IS curve is negatively sloped

- ▶ A fall in the interest rate motivates firms to increase investment spending, which drives up total planned spending (PE).
- ▶ To restore equilibrium in the goods market, output (a.k.a. actual expenditure, Y) must increase.

Fiscal policy and the IS curve

- ▶ We can use the $IS-LM$ model to see how fiscal policy (G and T) affects aggregate demand and output.
- ▶ Let's start by using the Keynesian cross to see how fiscal policy shifts the IS curve...

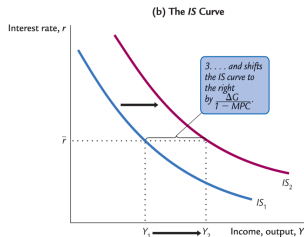
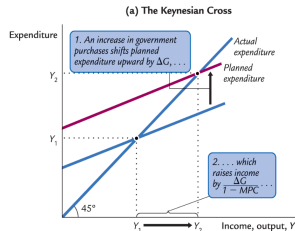
Shifting the IS curve: ΔG

At any value of r ,
 $\uparrow G \rightarrow \uparrow PE \rightarrow \uparrow Y$

...so the IS curve shifts to the right.

The horizontal distance of the IS shift equals

$$\Delta Y = \frac{1}{1 - MPC} \Delta G$$



NOW YOU TRY: Shifting the IS curve: ΔT

- ▶ Use the diagram of the Keynesian cross or loanable funds model to show how an increase in taxes shifts the IS curve.
- ▶ If you can, determine the size of the shift.

NOW YOU TRY: Shifting the IS curve: ΔT , answer

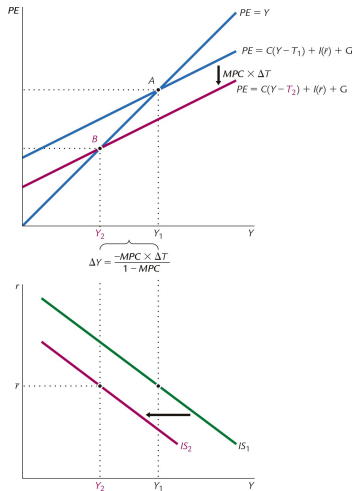
At any value of r ,

$\uparrow T \rightarrow \downarrow C \rightarrow \downarrow PE$

...so the IS curve shifts to the left.

The horizontal distance of the IS curve shift equals

$$\Delta Y = \frac{-MPC}{1 - MPC} \Delta T$$



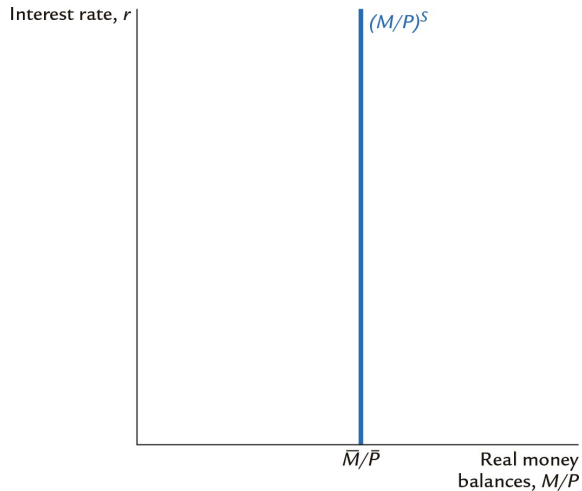
The theory of liquidity preference

- ▶ Due to John Maynard Keynes
- ▶ A simple theory in which the interest rate is determined by money supply and money demand

Money supply

The supply of real money balances is fixed:

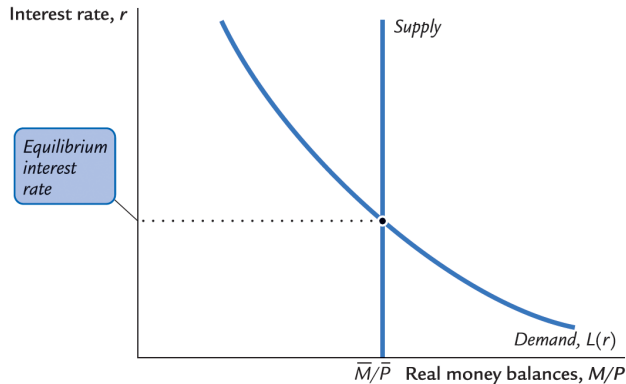
$$(M/P)^s = \bar{M}/\bar{P}$$



Money demand

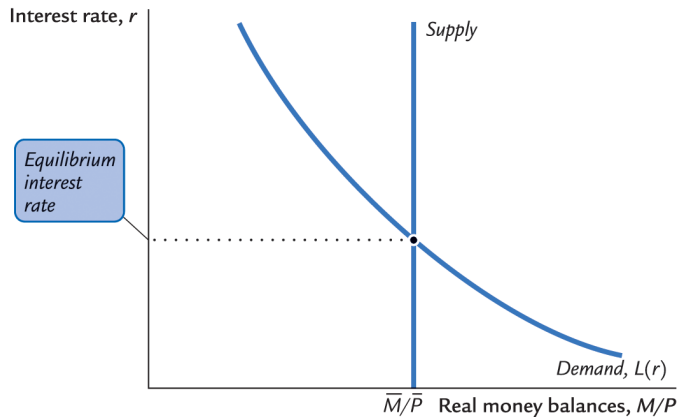
Demand for real money balances:

$$(M/P)^d = L(r)$$



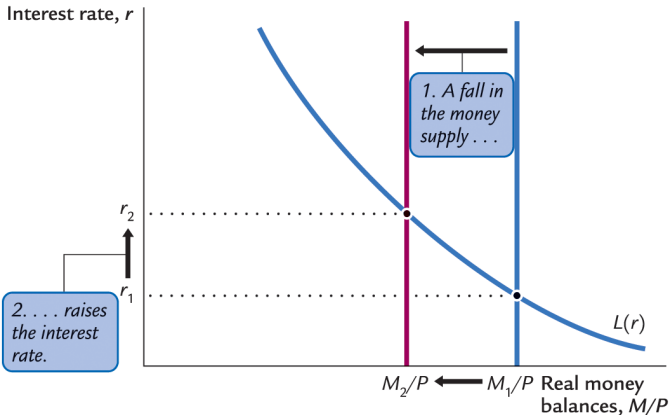
Equilibrium

Resulting in an
equilibrium interest rate
at r^*



How the Fed raises the interest rate

A fall in the money supply shifts MS in, which raises the interest rate.



CASE STUDY: Monetary tightening and interest rates, part 1

- ▶ Late 1970s: $\pi > 10$ percent
- ▶ October 1979: Fed Chair Paul Volcker announces that monetary policy would aim to reduce inflation.
- ▶ August 1979–April 1980: Fed reduces M/P 8.0 percent.
- ▶ January 1983: $\pi = 3.7$ percent

How do you think this policy change would affect nominal interest rates?

CASE STUDY: Monetary tightening and interest rates, part 2

The effects of a monetary tightening on nominal interest rates

	Short run	Long run
<i>Model</i>	Liquidity preference (<i>Keynesian</i>)	Quantity theory, Fisher effect (<i>classical</i>)
<i>Prices</i>	Sticky	Flexible
<i>Prediction</i>	$\Delta i > 0$	$\Delta i < 0$
<i>Actual outcome</i>	8/1979: $i = 10.4\%$ 4/1980: $i = 15.8\%$	8/1979: $i = 10.4\%$ 1/1983: $i = 8.2\%$

The LM curve

Now let's put Y back into the money demand function:

$$(M/P)^d = L(r, Y)$$

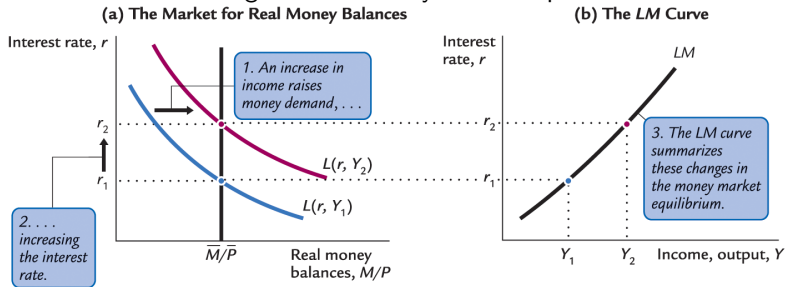
The **LM curve** is a graph of all combinations of r and Y that equate the supply and demand for real money balances.

The equation for the LM curve is:

$$\bar{M}/\bar{P} = L(r, Y)$$

Deriving the LM curve

As income increases from Y_1 to Y_2 , money demand shifts out, increasing the interest rate. The LM curve summarizes these changes in the money-market equilibrium.



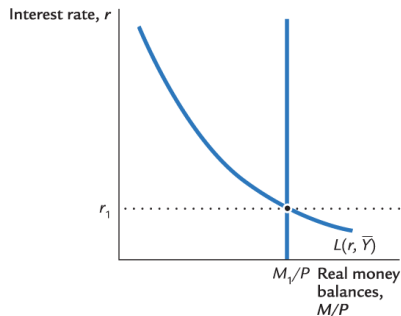
Why the LM curve is upward sloping

- ▶ An increase in income raises money demand.
- ▶ Since the supply of real balances is fixed, there is now excess demand in the money market at the initial interest rate.
- ▶ The interest rate must rise to restore equilibrium in the money market.

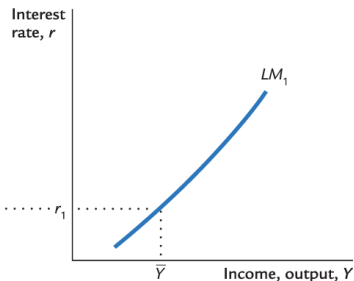
How ΔM shifts the LM curve

The Fed reduces the money from M_1 to M_2 , causing interest rates to rise. This then results in the LM curve shifting up to LM_2 .

(a) The Market for Real Money Balances



(b) The LM Curve

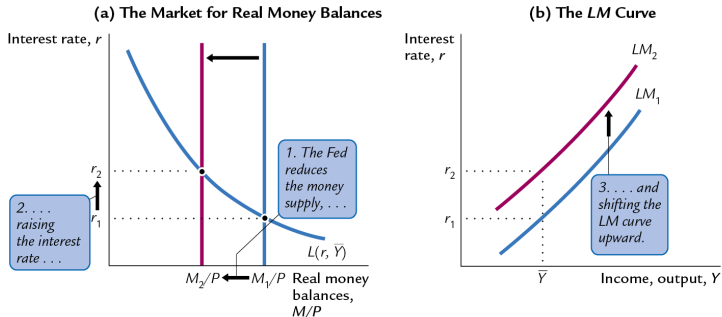


NOW YOU TRY: Shifting the LM curve

- ▶ Due to safety precautions during Covid-19, companies requested/required customers not to use cash for their payments.
- ▶ Use the liquidity preference model to show how these events shift the LM curve.

NOW YOU TRY: Shifting the LM curve, answer

Consumers use cash less frequently, and money demand will decrease from $L_1(r, Y)$ to $L_2(r, Y)$, causing interest rates to fall. This then results in the LM curve shifting up to LM_2 .

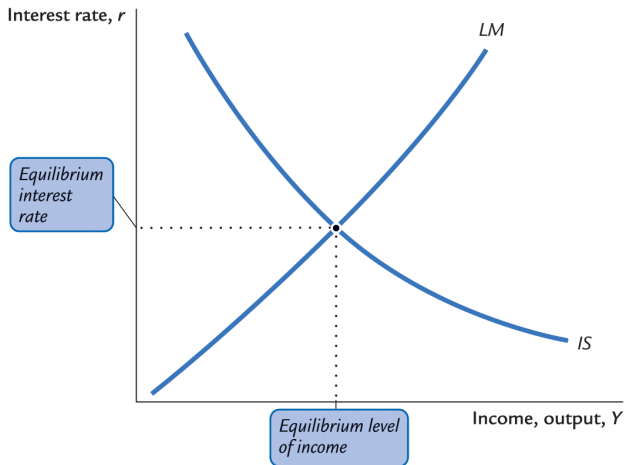


The short-run equilibrium

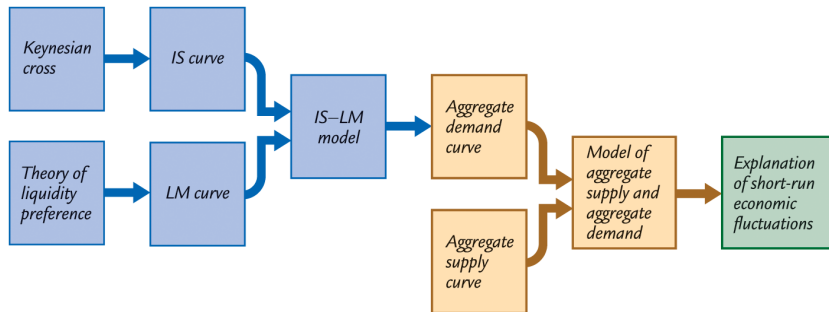
The short-run equilibrium is the combination of r and Y that simultaneously satisfies the equilibrium conditions in the goods and money markets:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}$$

$$\bar{M}/\bar{P} = L(r, Y)$$



The big picture



Preview of Chapter 14

In Chapter 14, we will

- ▶ use the $IS-LM$ model to analyze the impact of policies and shocks.
- ▶ learn how the aggregate demand curve comes from $IS-LM$.
- ▶ use the $IS-LM$ and $AD-AS$ models together to analyze the short-run and long-run effects of shocks.

Chapter Summary, Part 1

- ▶ Keynesian cross
 - ▶ basic model of income determination
 - ▶ takes fiscal policy and investment as exogenous
 - ▶ fiscal policy has a multiplier effect on income
- ▶ *IS* curve
 - ▶ comes from Keynesian cross when planned investment depends negatively on interest rate
 - ▶ shows all combinations of r and Y that equate planned expenditure with actual expenditure on goods and services

Chapter Summary, Part 2

- ▶ Theory of liquidity preference
 - ▶ basic model of interest rate determination
 - ▶ takes money supply and price level as exogenous
 - ▶ an increase in the money supply lowers the interest rate
- ▶ *LM* curve
 - ▶ comes from liquidity preference theory when money demand depends positively on income
 - ▶ shows all combinations of r and Y that equate demand for real money balances with supply

Chapter Summary, Part 3

- ▶ *IS-LM* model
 - ▶ The intersection of the *IS* and *LM* curves shows the unique point (Y, r) that satisfies equilibrium in both the goods and money markets.

Appendix: Keynesian cross multiplier — worked example

Setup. Consumption function $C = 100 + 0.6(Y - T)$, so $MPC = 0.6$. Investment: $I = 300$ (fixed for now). Fiscal: $G = 200$, $T = 100$.

Equilibrium $Y = C + I + G$:

$$Y = 100 + 0.6(Y - 100) + 300 + 200$$

$$Y = 540 + 0.6Y$$

$$0.4Y = 540 \implies Y^* = 1350.$$

Multiplier check. $\Delta G = +100$: $Y = 640 + 0.6Y$, so $Y^* = 1600$.

$$\frac{\Delta Y}{\Delta G} = \frac{250}{100} = 2.5 = \frac{1}{1 - 0.6} \cdot \checkmark$$

Tax multiplier. $\Delta T = -100$: adds $0.6 \cdot 100 = 60$ to autonomous spending.
 $\Delta Y = 60 \times 2.5 = 150$. Tax multiplier $= -MPC/(1 - MPC) = -1.5$.

[◀ back](#)

Appendix — deriving the IS curve algebraically

Now let investment depend on r : $I(r) = 300 - 20r$ (with r in %). Equilibrium: $Y = C + I(r) + G$:

$$Y = 100 + 0.6(Y - 100) + (300 - 20r) + 200$$
$$0.4Y = 540 - 20r$$

$$Y^S = 1350 - 50r.$$

(intercept $1350 = \frac{1}{0.4} \cdot 540$, slope $-\frac{1}{0.4} \cdot 20 = -50$)

Interpretation: higher $r \Rightarrow$ lower $I \Rightarrow$ lower Y — downward-sloping in (Y, r) space.

◀ back

Appendix — solving IS–LM equilibrium

Money market. $M/P = 400$; money demand $L(r, Y) = 0.4 Y - 50 r$.

$$400 = 0.4 Y - 50 r$$

$$Y^{LM} = 1000 + 125 r.$$

Equilibrium ($Y^{IS} = Y^{LM}$):

$$1350 - 50 r = 1000 + 125 r$$

$$350 = 175 r \implies r^* = 2\%, \quad Y^* = 1250.$$

Fiscal shock $\Delta G = +100$. IS shifts to $Y = 1600 - 50r$.

$$1600 - 50r = 1000 + 125r \implies r^* \approx 3.43\%,$$

$$Y^* \approx 1428.5.$$

So $\Delta Y \approx +178.5$, less than the pure Keynesian-cross prediction of $+250$: crowding out via higher r .